

1. Purpose

This document describes the functional behaviour of the Business Insights 360 solution — what the system does, how users interact with it, and what each component is expected to deliver. It is intended for the analytics team building the solution and for stakeholders reviewing it during UAT.

Where the BRD answers 'what the business needs', the FRD answers 'what the system must do'. It covers both the SQL reporting layer and the Power BI dashboard.

2. Functional Overview

The solution has two functional layers:

The SQL layer sits at the data source. It provides structured, filterable reports for product owners who need specific extracts — P&L summaries, top/bottom performance tables, and supply chain forecast accuracy — without interacting with a dashboard. Stored procedures handle the complexity, exposing simple filter parameters to the end user.

The Power BI layer is the primary interface for all other stakeholders. Five dashboard pages — Finance, Sales, Marketing, Supply Chain, and Executive — are each designed for a specific audience. The data behind all five pages flows from a single governed model, ensuring metric consistency across views. Users interact through slicers, dropdowns, toggle buttons, and drill-through capability. The report is published on Power BI Service and refreshes automatically via a gateway connected to MySQL and Excel.

3. Data Sources

Source	Type	Connection	Contents
MySQL Database	Relational DB	Direct connector	Fact tables: Sales, Forecast. Dim tables: Customer, Product, Market, Date
Excel — Targets	Flat file	Gateway (local)	Annual sales and margin targets by market and product
Excel — Market Share	Flat file	Gateway (local)	External market share data for AtliQ and competitors by sub-zone

Excel — Operational Costs	Flat file	Gateway (local)	Operational expense data used in Net Profit calculations
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4. Data Preparation (Power Query / SQL)

4.1 SQL Layer

- Source tables loaded into MySQL; schema designed in Snowflake format with ERD defining relationships between fact and dimension tables.
- Key SQL queries written for: Gross Sales to Net Profit P&L breakdown, top/bottom N products and customers by Net Sales, and Forecast Accuracy % with Net Error by customer and product.
- Stored procedures parameterised by fiscal year, market, and product segment — allowing product owners to filter without writing SQL.
- All queries validated against benchmark values before handoff.

4.2 Power Query (ETL)

- Data imported from MySQL via native Power BI connector and from Excel files via file path with gateway.
- Column names standardised across all tables; unnecessary columns removed before model load.
- Null handling applied: missing forecast quantities default to zero; missing customer or product references flagged for review.
- A dim_date table generated in Power Query to support time intelligence (fiscal year, quarter, month).
- Pre-invoice and post-invoice deduction tables merged with the sales fact table to enable net sales calculation at the row level.
- All transformations applied in Power Query before data enters the model — no transformation logic in DAX.

5. Data Model

The data model follows a Snowflake schema. Central fact tables — `fact_sales_monthly` and `fact_forecast_monthly` — connect to multiple dimension tables via foreign keys.

Table	Type	Key Fields / Role
fact_sales_monthly	Fact	customer_code, product_code, fiscal_year, sold_quantity, gross_price, pre_invoice_pct, post_invoice_discount_pct
fact_forecast_monthly	Fact	customer_code, product_code, fiscal_year, forecast_quantity
dim_customer	Dimension	customer code, customer name, market, sub zone, channel, platform
dim_product	Dimension	product_code, product, segment, category, division
dim_market	Dimension	market, sub zone, region
dim_date	Dimension	date, fiscal_year, fiscal_month, quarter (generated in Power Query)
fact_gross_price	Reference	product_code, fiscal_year, gross_price (joined to fact_sales)
fact_manufacturing_cost	Reference	product_code, fiscal_year, manufacturing_cost (COGS component)
fact_freight_cost	Reference	market, fiscal_year, freight_pct (COGS component)

All relationships are one-to-many from dimension to fact. No bidirectional filters are applied except where required for specific DAX calculations.

6. KPIs and DAX Measures

All KPIs are built as DAX measures. Calculated columns are used only where row-level context is required and cannot be replicated in a measure.

Measure	Used In	Logic Summary
Gross Sales	Finance, Sales	SUM(sold_quantity × gross_price) from fact_sales_monthly
Pre-Invoice Deduction	Finance, Sales	Gross Sales × average pre_invoice_pct from deductions table
Net Invoice Sales	Finance, Sales	Gross Sales minus Pre-Invoice Deduction
Post-Invoice Discount	Finance, Sales	Net Invoice Sales × average post_invoice_discount_pct
Net Sales	All views	Net Invoice Sales minus Post-Invoice Discount
COGS	Finance, Sales	SUM(manufacturing_cost + freight_cost) per product × sold_quantity
Gross Margin	Finance, Sales	Net Sales minus COGS
Gross Margin %	All views	DIVIDE(Gross Margin, Net Sales)
Operational Expenses	Marketing	Sourced from Excel opex file; joined by market and fiscal year
Net Profit	Finance, Marketing	Gross Margin minus Operational Expenses
Net Profit %	Finance, Marketing	DIVIDE(Net Profit, Net Sales)
Forecast Quantity	Supply Chain	SUM(forecast_quantity) from fact_forecast_monthly
Net Error	Supply Chain	Forecast Quantity minus Actual Sales Quantity
ABS Error	Supply Chain	SUMX of ABS(Net Error) at row level
Forecast Accuracy %	Supply Chain	1 minus DIVIDE(ABS Error, Forecast Quantity)
BM Net Sales / GM% / NP%	Finance	Selected benchmark (LY or Target) via SWITCH on slicer value
Market Share %	Executive	AtliQ Net Sales divided by total market revenue from external Excel file

7. Filters and User Interactions

All slicers and filters are persistent within a page session. Cross-page filters are not applied by default — each view is independent unless a drill-through is used.

Filter	Type	Pages Applied	Behaviour
Fiscal Year	Dropdown slicer	All pages	Filters all visuals on the selected page to the chosen fiscal year
Market / Region	Dropdown slicer	Finance, Sales, Executive	Filters data to selected market or region; multi-select supported
Customer	Dropdown slicer	Sales	Filters performance visuals to selected customer(s)
Product / Segment	Dropdown slicer	Sales, Marketing	Filters product and segment visuals accordingly
Benchmark Toggle	Button slicer	Finance	Switches benchmark metric between Last Year and Target across all Finance visuals
GM% / NP% Toggle	Toggle button	Marketing	Switches Y-axis of performance matrix between Gross Margin % and Net Profit %
Top / Bottom N	Numeric slicer	Finance, Sales	Controls how many top or bottom records appear in ranking visuals
Sales Trend Tooltip	Hover interaction	Sales	Surfaces a customer-specific NS and GM% trend on hover; no click required

8. Dashboard Pages

Each page is built for a specific stakeholder group. The design follows a consistent layout: KPI cards at the top, primary analysis visual in the centre, and supporting breakdown below.

Finance View	
Purpose	Gives the Finance team a live P&L summary from Gross Sales to Net Profit, with benchmark comparison against either Last Year or Target.
KPIs	Net Sales, Gross Margin %, Net Profit % (vs BM). All three show absolute value and % change vs benchmark.
Visuals	P&L statement table (matrix visual), KPI cards with conditional formatting, line chart (Net Sales trend vs BM), bar charts (Top/Bottom Products and Customers by Net Sales).
Questions Answered	Are we hitting our Net Sales and Gross Margin targets this year? Which markets or products are dragging our Net Profit? How is performance trending vs last year?

Sales View	
Purpose	Gives the Sales team visibility into customer and product-level profitability, with unit economics showing where revenue is lost through discounts and COGS.
KPIs	Net Sales, Gross Margin, Gross Margin % — at customer and product level.
Visuals	Waterfall or donut charts (unit economics: Gross Sales → Pre-Invoice Deduction → Net Invoice → Post-Invoice Discount → Net Sales → COGS → GM), table matrix (Customer/Product vs NS, GM, GM%), scatter plot (Performance Matrix: NS vs GM% by Market/Customer/Region), tooltip chart (NS and GM% trend per customer).
Questions Answered	Which customers are the most and least profitable? How much are discounts eroding our Net Sales? Which products have the highest and lowest Gross Margin %?

Marketing View	
Purpose	Helps the Marketing team understand how operational spending affects Gross Margin and Net Profit, and compare performance across segments and categories.
KPIs	Net Profit %, Gross Margin % — at segment, category, and product level.
Visuals	Waterfall chart (GM minus Opex = Net Profit), scatter/matrix visual (performance by Segment/Category/Product; toggleable between GM% and NP% on Y-axis), supporting tables for drill-down.
Questions Answered	Which product segments are actually profitable after operational costs? Is marketing spend improving or hurting our Net Profit %? Which categories should we prioritise or pull back from?

Supply Chain View

Purpose	Gives the Supply Chain team a forward-looking view of forecast accuracy and inventory risk, so issues can be addressed before they become stockouts or overstock situations.
KPIs	Forecast Accuracy %, Net Error, ABS Error (all vs Last Year). Risk flag per customer and product.
Visuals	KPI cards (FA%, Net Error, ABS Error with LY comparison), line chart (FA% vs Net Error trend over time), tables (Customer and Product detail: FA%, FA% LY, Net Error, Net Error%, Risk Factor), conditional colour-coding for risk classification.
Questions Answered	Which customers or products have the worst forecast accuracy? Where are we consistently over- or under-forecasting? Which items are at risk of stockout or excess inventory right now?

Executive View

Purpose	Provides top management with a single-page summary of company performance: market share, revenue mix, and top performers — without requiring drill-down into individual department views.
KPIs	Net Sales, Gross Margin %, Net Profit %, Forecast Accuracy %, Market Share % (all at company level).
Visuals	Line chart (AtliQ market share trend vs competitors), donut/bar charts (Revenue by Division and by Channel), bar tables (Top 5 Products and Top 5 Customers by Revenue), map or table (Sub-Zone revenue contribution %).
Questions Answered	How is AtliQ performing vs competitors in market share? Which divisions and channels are driving revenue? Who are our top customers and products this year? Which sub-zones are underperforming?

9. Functional Requirements

9.1 Data Pipeline

Sr. No.	Functional Requirement	Type
1.	Power BI must connect to MySQL via native connector and to Excel files via a configured data gateway.	Must Have
2.	All data transformations — column renaming, null handling, table merges, derived fields — must be completed in Power Query before data enters the model.	Must Have
3.	A dim date table must be generated in Power Query covering all fiscal years in the dataset, with fields for fiscal year, quarter, and month.	Must Have
4.	Pre-invoice and post-invoice deduction tables must be merged with the sales fact at row level, not aggregated upstream.	Must Have
5.	The gateway must be configured for automatic refresh so the dashboard always reflects current source data without manual intervention.	Must Have

9.2 Data Model

Sr. No.	Functional Requirement	Type
1.	The data model must follow a Snowflake schema with fact tables at the centre and dimension tables connected via one-to-many relationships.	Must Have
2.	No bidirectional relationships may be set by default. Exceptions require documented justification.	Must Have
3.	All KPI calculations must be built as DAX measures, not calculated columns, unless row-level context is strictly necessary.	Must Have
4.	A single benchmark measure must drive all benchmark comparisons in the Finance view, controlled by a user slicer — not hardcoded.	Must Have

9.3 Finance View

FR ID	Functional Requirement	Type
1.	The P&L matrix must display all line items from Gross Sales to Net Profit in the correct hierarchical order, with a benchmark column and % change column.	Must Have
2.	KPI cards for Net Sales, GM%, and NP% must show the current value, benchmark value, and directional indicator (up/down vs BM).	Must Have
3.	The benchmark slicer must dynamically switch all Finance visuals between Last Year and Target without requiring a page refresh.	Must Have

4.	The Net Sales trend line chart must plot the current year alongside the selected benchmark across all fiscal months.	Must Have
5.	Top/Bottom N product and customer charts must respond to a numeric slicer controlling the number of records shown.	Should Have

9.4 Sales View

Sr. No.	Functional Requirement	Type
1.	Unit economics visuals must show the step-by-step reduction from Gross Sales to Net Sales (pre- and post-invoice deductions) and then to Gross Margin (COGS).	Must Have
2.	Customer and product performance tables must support sorting by Net Sales, Gross Margin, and Gross Margin % independently.	Must Have
3.	The performance matrix scatter plot must allow the user to switch the dimension between Market, Customer, and Region using a slicer or button.	Should Have
4.	A tooltip chart showing NS and GM% trend must appear on hover over any customer row in the customer table. It must not require a click to activate.	Could Have

9.5 Marketing View

Sr. No.	Functional Requirement	Type
1.	The waterfall or bridge visual must show the path from Gross Margin to Net Profit by subtracting operational expense categories.	Must Have
2.	The performance matrix must include a toggle button that switches the Y-axis metric between Gross Margin % and Net Profit % without reloading the page.	Must Have
3.	Performance data must be viewable at Segment, Category, and Product levels via a hierarchy or slicer.	Should Have

9.6 Supply Chain View

Sr. No.	Functional Requirement	Type
1.	KPI cards for FA%, Net Error, and ABS Error must display current values alongside Last Year values with directional indicators.	Must Have
2.	Risk classification logic must be applied to every customer and product row: Excess Inventory where Net Error is materially positive; Out of Stock where Net Error is materially negative.	Must Have
3.	The FA% vs Net Error trend chart must plot both metrics on a dual-axis line chart by fiscal month.	Must Have

4.	Customer and product detail tables must include FA%, FA% LY, Net Error, Net Error %, and Risk Factor columns and support sorting by each.	Should Have
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9.7 Executive View

Sr. No.	Functional Requirement	Type
1.	The market share trend line must show AtliQ and at least two competitor groups over the selected fiscal year range.	Must Have
2.	Revenue breakdown by Division and by Channel must be shown using donut or bar charts, clearly labelled with % contribution.	Must Have
3.	Top 5 Products and Top 5 Customers by Revenue must be displayed as bar charts or ranked tables with NS and GM% values visible.	Must Have
4.	Sub-zone analysis must show revenue contribution % per sub-zone alongside at least one performance KPI (GM% or NP%).	Should Have

9.8 SQL Layer

Sr. No.	Functional Requirement	Type
1.	All SQL queries must be validated against Power BI DAX measures to confirm metric alignment before sign-off.	Must Have
2.	Stored procedures must accept parameters for at minimum: fiscal year, market, and product segment. Default values must be defined for each parameter.	Must Have
3.	The P&L stored procedure must output all line items from Gross Sales to Net Profit in a single result set with benchmark comparison columns.	Must Have
4.	The Forecast Accuracy stored procedure must return FA%, Net Error, ABS Error, and Risk classification per customer and per product.	Must Have

10. Assumptions

- The fiscal year runs April to March. All time intelligence in Power Query and DAX is built on this assumption. Any change to the fiscal calendar requires model updates.
- Operational expense data is provided at market and fiscal year granularity only. Product-level opex allocation is not possible with the current data and is therefore not in scope.
- Market share data from the external Excel file is taken at face value. The analytics team is not responsible for verifying the accuracy of competitor figures.
- All DAX measures are built to handle blank or zero denominator scenarios gracefully using DIVIDE() with an alternate result of 0 or blank, as appropriate.
- Dashboard performance will be acceptable for the volume of data in scope. If data volume significantly increases in future, query folding and aggregation optimisations may be needed.
- The benchmark slicer defaults to no selection. Users must make an active choice before benchmark comparisons populate.

11. Future Enhancements

The following items were discussed during stakeholder reviews but deferred to a future phase. They are noted here to preserve context for the next iteration:

Enhancement	Description
Row-Level Security (RLS)	Restrict data visibility by region or market based on the logged-in user's role. Relevant as the dashboard is rolled out to regional managers.
Automated Anomaly Alerts	Email or Teams notification when a KPI breaches a defined threshold — e.g. FA% drops below 70% or Net Profit % goes negative.
What-If Price Analysis	A parameter-driven simulation allowing Finance to model the impact of a price change on Gross Margin and Net Profit without altering source data.
Drill-Through to Transaction Detail	From any customer or product summary, allow the user to drill through to the underlying transaction rows for that record.
Mobile Layout	A mobile-optimised report layout for the Executive View to support leadership access on phones and tablets.
Competitor Expansion	Benchmarking Incorporate additional external data sources to expand market share analysis beyond the current Excel file.